

Commerce XI Maths App

Exercise 1.2

Detailed Answers with Explanations and Steps

Mathematics & Statistics, Part 1, Unit 1 - Sets and Relations | Total questions: 30

Question 1 - Q1

If $(x - 1, y + 4) = (1, 2)$, find the values of x and y .

Options

A. $x = 2, y = -2$

B. $x = 0, y = 6$

C. $x = 2, y = 6$

D. $x = 0, y = -2$

Correct answer: $x = 2, y = -2$

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives $x = 2, y = -2$. The other choices omit a required element, include an invalid element, or use a different operation.

Question 2 - Q2

If $(x + 1/3, y/3 - 1) = (1/3, 3/2)$, find x and y .

Options

A. $x = 0, y = 15/2$

B. $x = 2/3, y = 15/2$

C. $x = 0, y = 5/2$

D. $x = 2/3, y = 5/2$

Correct answer: $x = 0, y = 15/2$

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives $x = 0, y = 15/2$. The other choices omit a required element, include an invalid element, or use a different operation.

Question 3 - Q3(i)

If $A = \{a, b, c\}$ and $B = \{x, y\}$, find $A \times B$.

Options

A. $\{(a, x), (a, y), (b, x), (b, y), (c, x), (c, y)\}$

B. $\{(x, a), (x, b), (x, c), (y, a), (y, b), (y, c)\}$

C. $\{(a, a), (a, b), (a, c), (b, a), (b, b), (b, c), (c, a), (c, b), (c, c)\}$

D. $\{(x, x), (x, y), (y, x), (y, y)\}$

Correct answer: $\{(a, x), (a, y), (b, x), (b, y), (c, x), (c, y)\}$

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives $\{(a, x), (a, y), (b, x), (b, y), (c, x), (c, y)\}$. The other choices omit a required element, include an invalid element, or use a different operation.

Question 4 - Q3(ii)

If $A = \{a, b, c\}$ and $B = \{x, y\}$, find $B \times A$.

Options

- A. $\{(x, a), (x, b), (x, c), (y, a), (y, b), (y, c)\}$
- B. $\{(a, x), (a, y), (b, x), (b, y), (c, x), (c, y)\}$
- C. $\{(x, x), (x, y), (y, x), (y, y)\}$
- D. $\{(a, a), (b, b), (c, c)\}$

Correct answer: $\{(x, a), (x, b), (x, c), (y, a), (y, b), (y, c)\}$

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives $\{(x, a), (x, b), (x, c), (y, a), (y, b), (y, c)\}$. The other choices omit a required element, include an invalid element, or use a different operation.

Question 5 - Q3(iii)

If $A = \{a, b, c\}$ and $B = \{x, y\}$, find $A \times A$.

Options

- A. $\{(a, a), (a, b), (a, c), (b, a), (b, b), (b, c), (c, a), (c, b), (c, c)\}$
- B. $\{(a, x), (a, y), (b, x), (b, y), (c, x), (c, y)\}$
- C. $\{(a, a), (b, b), (c, c)\}$
- D. $\{(a, b), (a, c), (b, a), (b, c), (c, a), (c, b)\}$

Correct answer: $\{(a, a), (a, b), (a, c), (b, a), (b, b), (b, c), (c, a), (c, b), (c, c)\}$

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives $\{(a, a), (a, b), (a, c), (b, a), (b, b), (b, c), (c, a), (c, b), (c, c)\}$. The other choices omit a required element, include an invalid element, or use a different operation.

Question 6 - Q3(iv)

If $A = \{a, b, c\}$ and $B = \{x, y\}$, find $B \times B$.

Options

- A. $\{(x, x), (x, y), (y, x), (y, y)\}$
- B. $\{(x, y), (y, x)\}$
- C. $\{(x, x), (y, y)\}$
- D. $\{(a, x), (a, y), (b, x), (b, y), (c, x), (c, y)\}$

Correct answer: $\{(x, x), (x, y), (y, x), (y, y)\}$

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives $\{(x, x), (x, y), (y, x), (y, y)\}$. The other choices omit a required element, include an invalid element, or use a different operation.

Question 7 - Q4(i)

If $P = \{1, 2, 3\}$ and $Q = \{6, 4\}$, find $P \times Q$.

Options

- A. $\{(1, 6), (2, 6), (3, 6), (1, 4), (2, 4), (3, 4)\}$
- B. $\{(6, 1), (6, 2), (6, 3), (4, 1), (4, 2), (4, 3)\}$
- C. $\{(1, 6), (2, 6), (3, 6)\}$
- D. $\{(1, 4), (2, 4), (3, 4)\}$

Correct answer: $\{(1, 6), (2, 6), (3, 6), (1, 4), (2, 4), (3, 4)\}$

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives $\{(1, 6), (2, 6), (3, 6), (1, 4), (2, 4), (3, 4)\}$. The other choices omit a required element, include an invalid element, or use a different operation.

Question 8 - Q4(ii)

If $P = \{1, 2, 3\}$ and $Q = \{6, 4\}$, find $Q \times P$.

Options

- A. $\{(6, 1), (6, 2), (6, 3), (4, 1), (4, 2), (4, 3)\}$
- B. $\{(1, 6), (2, 6), (3, 6), (1, 4), (2, 4), (3, 4)\}$
- C. $\{(6, 1), (6, 2), (6, 3)\}$
- D. $\{(4, 1), (4, 2), (4, 3)\}$

Correct answer: $\{(6, 1), (6, 2), (6, 3), (4, 1), (4, 2), (4, 3)\}$

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives $\{(6, 1), (6, 2), (6, 3), (4, 1), (4, 2), (4, 3)\}$. The other choices omit a required element, include an invalid element, or use a different operation.

Question 9 - Q5(i)

Let $A = \{1, 2, 3, 4\}$, $B = \{4, 5, 6\}$, and $C = \{5, 6\}$.

Find $A \times (B \cap C)$.

Options

- A. $\{(1, 5), (1, 6), (2, 5), (2, 6), (3, 5), (3, 6), (4, 5), (4, 6)\}$
- B. $\{(1, 4), (2, 4), (3, 4), (4, 4)\}$
- C. $\{(1, 4), (1, 5), (1, 6), (2, 4), (2, 5), (2, 6), (3, 4), (3, 5), (3, 6), (4, 4), (4, 5), (4, 6)\}$
- D. $\{(4, 5), (4, 6)\}$

Correct answer: $\{(1, 5), (1, 6), (2, 5), (2, 6), (3, 5), (3, 6), (4, 5), (4, 6)\}$

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives $\{(1, 5), (1, 6), (2, 5), (2, 6), (3, 5), (3, 6), (4, 5), (4, 6)\}$. The other choices omit a required element, include an invalid element, or use a different operation.

Question 10 - Q5(ii)

Let $A = \{1, 2, 3, 4\}$, $B = \{4, 5, 6\}$, and $C = \{5, 6\}$.

Find $(A \times B) \cap (A \times C)$.

Options

- A. $\{(1, 5), (1, 6), (2, 5), (2, 6), (3, 5), (3, 6), (4, 5), (4, 6)\}$
- B. $\{(1, 4), (2, 4), (3, 4), (4, 4)\}$
- C. $\{(1, 4), (1, 5), (1, 6), (2, 4), (2, 5), (2, 6), (3, 4), (3, 5), (3, 6), (4, 4), (4, 5), (4, 6)\}$
- D. $\{(4, 4), (4, 5), (4, 6)\}$

Correct answer: $\{(1, 5), (1, 6), (2, 5), (2, 6), (3, 5), (3, 6), (4, 5), (4, 6)\}$

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives $\{(1, 5), (1, 6), (2, 5), (2, 6), (3, 5), (3, 6), (4, 5), (4, 6)\}$. The other choices omit a required element, include an invalid element, or use a different operation.

Question 11 - Q5(iii)

Let $A = \{1, 2, 3, 4\}$, $B = \{4, 5, 6\}$, and $C = \{5, 6\}$.

Find $A \times (B \cup C)$.

Options

- A. $\{(1, 4), (1, 5), (1, 6), (2, 4), (2, 5), (2, 6), (3, 4), (3, 5), (3, 6), (4, 4), (4, 5), (4, 6)\}$
- B. $\{(1, 5), (1, 6), (2, 5), (2, 6), (3, 5), (3, 6), (4, 5), (4, 6)\}$
- C. $\{(1, 4), (2, 4), (3, 4), (4, 4)\}$
- D. $\{(4, 4), (4, 5), (4, 6)\}$

Correct answer: $\{(1, 4), (1, 5), (1, 6), (2, 4), (2, 5), (2, 6), (3, 4), (3, 5), (3, 6), (4, 4), (4, 5), (4, 6)\}$

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives $\{(1, 4), (1, 5), (1, 6), (2, 4), (2, 5), (2, 6), (3, 4), (3, 5), (3, 6), (4, 4), (4, 5), (4, 6)\}$. The other choices omit a required element, include an invalid element, or use a different operation.

Question 12 - Q5(iv)

Let $A = \{1, 2, 3, 4\}$, $B = \{4, 5, 6\}$, and $C = \{5, 6\}$.

Find $(A \times B) \cup (A \times C)$.

Options

- A. $\{(1, 4), (1, 5), (1, 6), (2, 4), (2, 5), (2, 6), (3, 4), (3, 5), (3, 6), (4, 4), (4, 5), (4, 6)\}$
- B. $\{(1, 5), (1, 6), (2, 5), (2, 6), (3, 5), (3, 6), (4, 5), (4, 6)\}$
- C. $\{(1, 4), (2, 4), (3, 4), (4, 4)\}$
- D. $\{(4, 4), (4, 5), (4, 6)\}$

Correct answer: $\{(1, 4), (1, 5), (1, 6), (2, 4), (2, 5), (2, 6), (3, 4), (3, 5), (3, 6), (4, 4), (4, 5), (4, 6)\}$

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives $\{(1, 4), (1, 5), (1, 6), (2, 4), (2, 5), (2, 6), (3, 4), (3, 5), (3, 6), (4, 4), (4, 5), (4, 6)\}$. The other choices omit a required element, include an invalid element, or use a different operation.

Question 13 - Q6

Express $\{(x, y) / x^2 + y^2 = 100 \text{ where } x, y \in W\}$ as a set of ordered pairs.

Options

- A. $\{(0, 10), (6, 8), (8, 6), (10, 0)\}$
- B. $\{(6, 8), (8, 6)\}$
- C. $\{(0, 10), (10, 0)\}$
- D. $\{(0, 10), (6, 8), (8, 6), (10, 0), (10, 10)\}$

Correct answer: $\{(0, 10), (6, 8), (8, 6), (10, 0)\}$

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives $\{(0, 10), (6, 8), (8, 6), (10, 0)\}$. The other choices omit a required element, include an invalid element, or use a different operation.

Question 14 - Q7(i) Domain

Write the domain of the following relation.

$$\{(a, b) / a \in \mathbb{N}, a < 6 \text{ and } b = 4\}$$

Options

- A. $\{1, 2, 3, 4, 5\}$
- B. $\{0, 1, 2, 3, 4, 5\}$
- C. $\{1, 2, 3, 4, 5, 6\}$
- D. $\{4\}$

Correct answer: $\{1, 2, 3, 4, 5\}$

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives $\{1, 2, 3, 4, 5\}$. The other choices omit a required element, include an invalid element, or use a different operation.

Question 15 - Q7(i) Range

Write the range of the following relation.

$$\{(a, b) / a \in \mathbb{N}, a < 6 \text{ and } b = 4\}$$

Options

- A. $\{4\}$
- B. $\{1, 2, 3, 4, 5\}$
- C. $\{1, 2, 3, 4\}$
- D. $\{1, 2, 3, 4, 5, 6\}$

Correct answer: $\{4\}$

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives $\{4\}$. The other choices omit a required element, include an invalid element, or use a different operation.

Question 16 - Q7(ii) Domain

Write the domain of the following relation.

$$\{(a, b) / a, b \in \mathbb{N}, a + b = 12\}$$

Options

- A. {1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11}
- B. {0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12}
- C. {1, 2, 3, 4, 5, 6}
- D. {2, 3, 4, 5, 6, 7, 8, 9, 10}

Correct answer: {1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11}

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives {1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11}. The other choices omit a required element, include an invalid element, or use a different operation.

Question 17 - Q7(ii) Range

Write the range of the following relation.

$\{(a, b) / a, b \in \mathbb{N}, a + b = 12\}$

Options

- A. {11, 10, 9, 8, 7, 6, 5, 4, 3, 2, 1}
- B. {12, 11, 10, 9, 8, 7, 6, 5, 4, 3, 2, 1}
- C. {1, 2, 3, 4, 5, 6}
- D. {10, 9, 8, 7, 6, 5, 4, 3, 2}

Correct answer: {11, 10, 9, 8, 7, 6, 5, 4, 3, 2, 1}

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives {11, 10, 9, 8, 7, 6, 5, 4, 3, 2, 1}. The other choices omit a required element, include an invalid element, or use a different operation.

Question 18 - Q7(iii) Domain

Write the domain of the following relation.

$\{(2, 4), (2, 5), (2, 6), (2, 7)\}$

Options

- A. {2}
- B. {4, 5, 6, 7}
- C. {2, 4, 5, 6, 7}
- D. {2, 2, 2, 2}

Correct answer: {2}

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives $\{2\}$. The other choices omit a required element, include an invalid element, or use a different operation.

Question 19 - Q7(iii) Range

Write the range of the following relation.

$\{(2, 4), (2, 5), (2, 6), (2, 7)\}$

Options

A. $\{4, 5, 6, 7\}$

B. $\{2\}$

C. $\{2, 4, 5, 6, 7\}$

D. $\{4, 5, 6\}$

Correct answer: $\{4, 5, 6, 7\}$

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives $\{4, 5, 6, 7\}$. The other choices omit a required element, include an invalid element, or use a different operation.

Question 20 - Q8

Let $A = \{6, 8\}$ and $B = \{1, 3, 5\}$.

Let $R = \{(a, b) / a \in A, b \in B, a - b \text{ is an even number}\}$.

Which set shows that R is an empty relation from A to B ?

Options

A. $R = \{ \}$

B. $R = \{(6, 1), (6, 3), (6, 5)\}$

C. $R = \{(8, 1), (8, 3), (8, 5)\}$

D. $R = \{(6, 1), (8, 3)\}$

Correct answer: $R = \{ \}$

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives $R = \{ \}$. The other choices omit a required element, include an invalid element, or use a different operation.

Question 21 - Q9(i) Roster form

Write the relation in the Roster form.

$R_1 = \{(a, a^2) / a \text{ is prime number less than } 15\}$

Options

- A. $R_1 = \{(2, 4), (3, 9), (5, 25), (7, 49), (11, 121), (13, 169)\}$
- B. $R_1 = \{(2, 4), (3, 9), (5, 25), (7, 49), (11, 121)\}$
- C. $R_1 = \{(1, 1), (2, 4), (3, 9), (5, 25), (7, 49), (11, 121), (13, 169)\}$
- D. $R_1 = \{(2, 2), (3, 3), (5, 5), (7, 7), (11, 11), (13, 13)\}$

Correct answer: $R_1 = \{(2, 4), (3, 9), (5, 25), (7, 49), (11, 121), (13, 169)\}$

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives $R_1 = \{(2, 4), (3, 9), (5, 25), (7, 49), (11, 121), (13, 169)\}$. The other choices omit a required element, include an invalid element, or use a different operation.

Question 22 - Q9(i) Domain

Find the domain of the following relation.

$$R_1 = \{(a, a^2) / a \text{ is prime number less than } 15\}$$

Options

- A. $\{2, 3, 5, 7, 11, 13\}$
- B. $\{1, 2, 3, 5, 7, 11, 13\}$
- C. $\{4, 9, 25, 49, 121, 169\}$
- D. $\{2, 3, 5, 7, 11\}$

Correct answer: $\{2, 3, 5, 7, 11, 13\}$

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives $\{2, 3, 5, 7, 11, 13\}$. The other choices omit a required element, include an invalid element, or use a different operation.

Question 23 - Q9(i) Range

Find the range of the following relation.

$$R_1 = \{(a, a^2) / a \text{ is prime number less than } 15\}$$

Options

- A. $\{4, 9, 25, 49, 121, 169\}$
- B. $\{2, 3, 5, 7, 11, 13\}$
- C. $\{1, 4, 9, 25, 49, 121, 169\}$
- D. $\{4, 9, 25, 49, 121\}$

Correct answer: $\{4, 9, 25, 49, 121, 169\}$

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives {4, 9, 25, 49, 121, 169}. The other choices omit a required element, include an invalid element, or use a different operation.

Question 24 - Q9(ii) Roster form

Write the relation in the Roster form.

$$R_2 = \{(a, 1/a) / 0 < a \leq 5, a \in \mathbb{N}\}$$

Options

- A. $R_2 = \{(1, 1), (2, 1/2), (3, 1/3), (4, 1/4), (5, 1/5)\}$
- B. $R_2 = \{(1, 1), (2, 2), (3, 3), (4, 4), (5, 5)\}$
- C. $R_2 = \{(1, 1), (2, 1/2), (3, 1/3), (4, 1/4)\}$
- D. $R_2 = \{(0, 0), (1, 1), (2, 1/2), (3, 1/3), (4, 1/4), (5, 1/5)\}$

Correct answer: $R_2 = \{(1, 1), (2, 1/2), (3, 1/3), (4, 1/4), (5, 1/5)\}$

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives $R_2 = \{(1, 1), (2, 1/2), (3, 1/3), (4, 1/4), (5, 1/5)\}$. The other choices omit a required element, include an invalid element, or use a different operation.

Question 25 - Q9(ii) Domain

Find the domain of the following relation.

$$R_2 = \{(a, 1/a) / 0 < a \leq 5, a \in \mathbb{N}\}$$

Options

- A. {1, 2, 3, 4, 5}
- B. {0, 1, 2, 3, 4, 5}
- C. {1, 1/2, 1/3, 1/4, 1/5}
- D. {1, 2, 3, 4}

Correct answer: {1, 2, 3, 4, 5}

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives {1, 2, 3, 4, 5}. The other choices omit a required element, include an invalid element, or use a different operation.

Question 26 - Q9(ii) Range

Find the range of the following relation.

$$R_2 = \{(a, 1/a) / 0 < a \leq 5, a \in \mathbb{N}\}$$

Options

A. $\{1, 1/2, 1/3, 1/4, 1/5\}$

B. $\{1, 2, 3, 4, 5\}$

C. $\{1/2, 1/3, 1/4, 1/5\}$

D. $\{0, 1, 1/2, 1/3, 1/4, 1/5\}$

Correct answer: $\{1, 1/2, 1/3, 1/4, 1/5\}$

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives $\{1, 1/2, 1/3, 1/4, 1/5\}$. The other choices omit a required element, include an invalid element, or use a different operation.

Question 27 - Q10

$R = \{(a, b) / b = a + 1, a \in \mathbb{Z}, 0 < a < 5\}$

Find the Range of R.

Options

A. $\{2, 3, 4, 5\}$

B. $\{1, 2, 3, 4\}$

C. $\{1, 2, 3, 4, 5\}$

D. $\{2, 3, 4\}$

Correct answer: $\{2, 3, 4, 5\}$

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives $\{2, 3, 4, 5\}$. The other choices omit a required element, include an invalid element, or use a different operation.

Question 28 - Q11(i)

Find the following relation as a set of ordered pairs.

$\{(x, y) / y = 3x, x \in \{1, 2, 3\}, y \in \{3, 6, 9, 12\}\}$

Options

A. $\{(1, 3), (2, 6), (3, 9)\}$

B. $\{(1, 3), (2, 6), (3, 9), (4, 12)\}$

C. $\{(3, 1), (6, 2), (9, 3)\}$

D. $\{(1, 3), (2, 6), (3, 12)\}$

Correct answer: $\{(1, 3), (2, 6), (3, 9)\}$

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives $\{(1, 3), (2, 6), (3, 9)\}$. The other choices omit a required element, include an invalid element, or use a different operation.

Question 29 - Q11(ii)

Find the following relation as a set of ordered pairs.

$$\{(x, y) / y > x + 1, x \in \{1, 2\} \text{ and } y \in \{2, 4, 6\}\}$$

Options

- A. $\{(1, 4), (1, 6), (2, 4), (2, 6)\}$
- B. $\{(1, 2), (1, 4), (1, 6), (2, 4), (2, 6)\}$
- C. $\{(1, 4), (2, 6)\}$
- D. $\{(1, 2), (2, 4)\}$

Correct answer: $\{(1, 4), (1, 6), (2, 4), (2, 6)\}$

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives $\{(1, 4), (1, 6), (2, 4), (2, 6)\}$. The other choices omit a required element, include an invalid element, or use a different operation.

Question 30 - Q11(iii)

Find the following relation as a set of ordered pairs.

$$\{(x, y) / x + y = 3, x, y \in \{0, 1, 2, 3\}\}$$

Options

- A. $\{(0, 3), (1, 2), (2, 1), (3, 0)\}$
- B. $\{(1, 2), (2, 1)\}$
- C. $\{(0, 3), (1, 2), (2, 1)\}$
- D. $\{(0, 3), (1, 2), (2, 1), (3, 0), (3, 3)\}$

Correct answer: $\{(0, 3), (1, 2), (2, 1), (3, 0)\}$

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives $\{(0, 3), (1, 2), (2, 1), (3, 0)\}$. The other choices omit a required element, include an invalid element, or use a different operation.